

Robust-reference profile correlation: pointwise inference and finite-sample limits

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Abstract

Paired observations may be associated through their magnitudes of departure from marginal references, even when signed association is not the scientific target. We study the correlation of absolute-deviation profiles generated by median/MAD-scaled Huber reference locations. Under fixed regular independent and identically distributed paired laws, we derive a complete influence-function expansion and consistent studentization, accounting jointly for reference fitting and profile moments. A shared-sign construction has independent profiles at the population references but can exhibit substantial fitted-profile bias and empirical null rejection distortion. A staged comparison separates correlated location errors that persist with population scale fixed from additional amplification under complete median/MAD fitting. A model-specific result identifies nonregular median estimation despite a nonzero population Huber slope. An effective local conditioning index helps organize empirical bridge transitions but does not provide a universal cutoff. A regular nonzero-effect skew example also exhibits slow convergence and undercoverage. The contribution is a pointwise inferential foundation and a worked account of finite-sample limits, rather than a generally calibrated finite-sample procedure.

Keywords: generated profiles; influence function; Huber location; median absolute deviation; weak null; finite-sample distortion.

1. Introduction

Provisional: the Introduction below retains the current author-discussion text pending Hoorn’s revision. The short literature paragraph added at its end is an integration input for author review.

Some paired-data questions concern whether the same observational units are unusual in two different domains. An observation far from its own group in one variable may also be far from its group in another variable, regardless of whether the two signed deviations have the same direction. This question differs from ordinary signed association, equality of marginal distributions, and general dependence.

The correlation-of-divergency construction of Hoorn (2025) approaches this question by comparing observation-level summaries of all-to-all within-group divergence. The present paper retains the paired-profile motivation but studies a different construction: distance from a fitted robust marginal reference. Replacing an all-to-all divergence profile with a reference-based profile changes the estimand; it is not simply a new computational formula for the original coefficient.

Our primary target is

$$\rho_P = \text{Corr}(|X - T_X|, |Y - T_Y|),$$

where T_X, T_Y are marginal Huber location functionals whose scales are determined by the median absolute deviation, or MAD. The references are estimated from the same paired observations used to construct the profiles. Consequently, the profiles are generated quantities, and their sampling uncertainty includes the effect of reference estimation.

Robust location estimation provides the relevant methodological background (Huber, 1964). However, a bounded location score does not make a subsequent correlation of unbounded distances globally robust. Nor does a robust reference ensure that its finite-sample estimate remains stable in all distributional geometries.

The terminology must also be distinguished from earlier absolute-deviation correlation measures. In particular, the interval-level coefficient of Stavig (1982) is based on an absolute discrepancy between signed standardized values, rather than the Pearson correlation of unsigned distances from two fitted robust references. We do not claim novelty for taking absolute values, using robust location ingredients, or applying the delta method. The contribution concerns the complete inference and finite-sample behavior of this particular generated-profile construction.

The paper addresses three linked questions. First, under which fixed regular IID conditions does complete influence-function inference apply? Second, how can linked reference errors generate misleading profile association when the population profile correlation is zero? Third, how much of the resulting transition can be organized by local conditioning of the nuisance equations?

These questions lead to deliberately limited conclusions. The regular theorem is pointwise, not uniform near degeneracy. Reference fitting can cause severe distortion, but does not always do so. The conditioning index is an explanatory quantity, not a universal cutoff or an operational validity gate.

Related robust correlation methods also use median/MAD ingredients. Shevlyakov and Vilchevski (2002) study robust estimation of signed bivariate correlation; their principal-variable constructions do not define the unsigned radius target studied here. General generated-regressor theory already accounts for first-stage estimation in influence functions (Hahn and Ridder, 2013). Our derivation is a problem-specific application of nuisance propagation, not a new general principle. The distinctions are therefore in the target, the complete median/MAD/Huber and paired five-moment calculation, and the worked finite-sample mechanism. A finite literature comparison does not establish global priority.

2. Profiles, estimands, and the original construction

2.1 Paired observations and reference functional

Let $Z = (X, Y) \sim P$ be one scientifically meaningful paired unit. Units are IID when applying the regular theorem. For a continuous margin W define

$$m_W = \text{Med}(W), \quad d_W = \text{Med}|W - m_W|, \quad s_W = kd_W, \quad E\psi_c\{(W - T_W)/s_W\} = 0,$$

where $\psi_c(u) = \max(-c, \min(u, c))$, $k = 1.4826$, and $c = 1.345$. The raw MAD d and scaled MAD s are different quantities. Both sample medians use the midpoint convention, including the median of absolute deviations. Selected roots, their uniqueness and the numerical approximation are treated in Appendix A and S1. The main profiles are uncapped: $a = |X - T_X|$ and $b = |Y - T_Y|$.

2.2 Primary and secondary effects

Write $\nu = E(ab)$, $\mu_a = E(a)$, $\mu_b = E(b)$, $q_a = E(a^2)$, $q_b = E(b^2)$, and $v_a = q_a - \mu_a^2$, $v_b = q_b - \mu_b^2$. At positive profile variances,

$$\rho_P = \frac{\nu - \mu_a\mu_b}{\sqrt{v_a v_b}}, \quad C = \frac{\nu}{\mu_a\mu_b} = 1 + \rho_P CV(a) CV(b).$$

Positive profile correlation means that units far from one reference tend also to be far from the other; negative correlation means that large and small reference distances tend to be paired. Zero is a weak covariance null, not independence of the raw pair. The target is neither marginal distribution equality, general dependence, signed association, nor complete dyadic geometry. C is secondary because it also weights association by marginal profile CVs. Both effects are invariant to separate positive rescaling of the profiles. The qualification is **robust reference, not globally robust correlation**.

For $\hat{a}_i = |X_i - \hat{T}_X|$ and $\hat{b}_i = |Y_i - \hat{T}_Y|$, use their ordinary sample Pearson correlation $\hat{\rho}_P$ and $\hat{C} = P_n(\hat{a}\hat{b})/(\bar{a}\bar{b})$. Zero empirical profile variance is undefined correlation, not zero association. Empirical CVs in identities below use divisor n , consistently with the moments.

2.3 Relation to all-to-all divergence

For this project's one-dimensional L2 implementation,

$$D_{X,i} = \left\{ \frac{1}{n-1} \sum_{j \neq i} (X_i - X_j)^2 \right\}^{1/2}, \quad D_{X,i}^2 = \frac{n}{n-1} \{ (X_i - \bar{X})^2 + s_{X,n}^2 \},$$

where $s_{X,n}^2 = n^{-1} \sum_i (X_i - \bar{X})^2$. The identity follows by expanding $X_i - X_j = (X_i - \bar{X}) - (X_j - \bar{X})$ and summing, since $\sum_j (X_j - \bar{X}) = 0$. This is an exact sample identity for the stated implementation. Its common variance floor distinguishes it from a simple reference distance. The robust-reference C uses different profiles from the original all-to-all coefficient. No assertion about whether a planned normalization change has appeared in the latest public preprint is needed for this distinction.

3. Complete influence-function inference

3.1 Reference and moment influences

Suppress the margin subscript and let $U = (W - T)/s$, $A = P(|U| < c)$, $B = E\{U1(|U| < c)\}$. At a regular law,

$$\begin{aligned} IF_m(w) &= \frac{1/2 - 1\{w \leq m\}}{f(m)}, \\ IF_d(w) &= \frac{1/2 - 1\{|w - m| \leq d\} - \{f(m+d) - f(m-d)\}IF_m(w)}{f(m+d) + f(m-d)}, \\ IF_T(w) &= \frac{s}{A} \psi_c\{(w - T)/s\} - \frac{B}{A} k IF_d(w). \end{aligned}$$

The last term includes the median's indirect contribution through MAD. At a symmetric population B is zero. Define $g_X = E \text{sign}(X - T_X)$, $h_X = E\{\text{sign}(X - T_X)b\}$, and analogously $g_Y = E \text{sign}(Y - T_Y)$, $h_Y = E\{\text{sign}(Y - T_Y)a\}$. Then

$$\begin{aligned} IF_\nu &= ab - \nu - h_X IF_{T_X} - h_Y IF_{T_Y}, \\ IF_{\mu_a} &= a - \mu_a - g_X IF_{T_X}, \quad IF_{\mu_b} = b - \mu_b - g_Y IF_{T_Y}, \\ IF_{q_a} &= a^2 - q_a - 2E(X - T_X)IF_{T_X}, \\ IF_{q_b} &= b^2 - q_b - 2E(Y - T_Y)IF_{T_Y}. \end{aligned}$$

All influences are evaluated on the same pair. In moment order $M = (\nu, \mu_a, \mu_b, q_a, q_b)^\top$, with $D = \sqrt{v_a v_b}$,

$$\nabla \rho_P = \begin{pmatrix} 1/D \\ -\mu_b/D + \rho_P \mu_a/v_a \\ -\mu_a/D + \rho_P \mu_b/v_b \\ -\rho_P/(2v_a) \\ -\rho_P/(2v_b) \end{pmatrix}, \quad IF_\rho = \nabla \rho_P^\top IF_M.$$

One influence value is contributed by each intact pair. Its variance retains numerator–denominator, direct–reference and cross-margin covariances. Shared pairwise edges cannot be treated as IID observations.

3.2 Pointwise theorem

Theorem 1. Under Conditions R1–R5 in Appendix A and the local class verification in S1.3, at a fixed regular law,

$$\sqrt{n}(\hat{\rho}_P - \rho_P) = n^{-1/2} \sum_i IF_\rho(Z_i) + o_P(1) \Rightarrow N(0, V_\rho), \quad V_\rho = E(IF_\rho^2) > 0.$$

The sample variance of complete plug-in influences, $\hat{V}_\rho = (n-1)^{-1} \sum_i (\widehat{IF}_{\rho,i} - \overline{\widehat{IF}_\rho})^2$, converges to V_ρ . Therefore $\sqrt{n}(\hat{\rho}_P - \rho_P)/\sqrt{\hat{V}_\rho} \Rightarrow N(0, 1)$. The Wald interval is $\hat{\rho}_P \pm z_{1-\alpha/2} \sqrt{\hat{V}_\rho/n}$.

Appendix A provides the proof route and S1 supplies its explicit Bahadur, paired-score, entropy, density, and studentization arguments. The theorem is **pointwise, not uniform**, and gives no minimum safe sample size. S1.7 separates its root-residual condition from the actual fixed-tolerance solver. At global independence the reference coefficients cancel; cancellation also occurs in some symmetric dependent profile-null laws, but not under every weak null. For general nonzero correlation the quadratic terms require the stronger moments in R4. The reduced C and weak-null arguments are in S1.6.

4. A worked finite-sample failure

4.1 Shared-sign construction

Let $X = S \exp(\tau U)$, $Y = S \exp(\tau V)$, $\tau > 0$, where S is symmetric Rademacher and U, V are independent standard normal variables independent of S .

Proposition 1. Both population medians and Huber references are zero, the raw MAD is one, and the population profiles are independent. Thus $\rho_P = 0, C = 1$, although X, Y are dependent. The marginal density at zero is zero. For the midpoint sample median, $\sqrt{n}|\hat{m}| \rightarrow \infty$ in probability, whereas the population Huber score slope is $-\Phi\{\log(ck)/\tau\}/k < 0$.

Appendix B proves the proposition. At $\tau = .10$ this slope is about $-.67449$; the example is not explained by a nearly flat population Huber equation. The positive median-density condition of the regular theorem fails.

4.2 Location error and additional fitting amplification

With population scale fixed, a Huber fit equals the sample mean whenever all fitted residuals are in its linear region. The sample means satisfy

$$E(\bar{X}\bar{Y}) = e^{\tau^2}/n, \quad E(\bar{X}^2) = E(\bar{Y}^2) = e^{2\tau^2}/n, \quad \text{Corr}(\bar{X}, \bar{Y}) = e^{-\tau^2}.$$

These follow from the intact-pair moments and independence across units. They exhibit a direct shared-sign location-error pathway with no estimated median or MAD. Full fitting can additionally move the median, change the MAD about that median, and amplify the location errors. Population symmetry makes B zero; such finite-sample scale propagation is not a nonzero first-order MAD derivative at that law. Section 6.4 compares these paths on identical samples.

For fixed offsets and independent radii independent of S, if $R_X > |t_X|, R_Y > |t_Y|$ almost surely, then $|SR_X - t_X| = R_X - St_X$, so the induced covariance is $t_X t_Y$ and the correlation is

$$\frac{t_X t_Y}{\sqrt{(v_X + t_X^2)(v_Y + t_Y^2)}}, \quad v_X = \text{Var}(R_X), \quad v_Y = \text{Var}(R_Y).$$

For lognormal radii this no-crossing condition does not hold globally for nonzero offsets. With fitted sample offsets the exact no-crossing sample identity, using divisor n, is instead

$$\begin{aligned} \text{Cov}_n(\hat{a}, \hat{b}) &= \text{Cov}_n(R_X, R_Y) - \hat{T}_X \text{Cov}_n(S, R_Y) \\ &\quad - \hat{T}_Y \text{Cov}_n(R_X, S) + \hat{T}_X \hat{T}_Y \text{Var}_n(S). \end{aligned}$$

The empirical cross terms cannot be discarded by inserting fitted offsets into the population identity. Conditional on root-n reference errors and an appropriate no-crossing approximation, radial variance of order τ^2 gives a product contribution of correlation scale $1/(n\tau^2)$. Comparing that with $n^{-1/2}$ suggests $\tau \asymp n^{-1/4}$. This is a **conditional rate heuristic**, not a local-to-degeneracy theorem for the complete estimator.

5. Effective local conditioning

For nuisance errors in raw-MAD units and the fixed score normalization in S1.2, the marginal derivative is

$$J = \begin{pmatrix} df(m) & 0 & 0 \\ d\{f(m+d) - f(m-d)\} & d\{f(m+d) + f(m-d)\} & 0 \\ 0 & -B & -A/k \end{pmatrix}, \quad I_n = \sqrt{n} \sigma_{\min}(J).$$

At a nonsingular fixed law the weakest standardized direction has a local linear amplification scale $1/I_n$. Its covariance also involves the score covariance $\Omega_W = E(\xi_W \xi_W^\top)$ explicitly defined in S1.2: $\text{Var}\{(\hat{\theta}_W - \theta_W)/d_W\} \approx J^{-1} \Omega_W J^{-\top}/n$. The target additionally depends on its nuisance projection, profile variation and direct-reference covariance. Matching J does not match the entire first-order distribution. Residual differences cannot automatically be classified as higher order.

In the unbridged model, $J = \text{diag}(0, 1/(\tau\sqrt{2\pi}), -A/k)$. Thus $I_n = 0$ for every positive τ ; it cannot distinguish the severe and diffuse controls. Separate positive-density bridge comparisons are now provided in S2, including all 24 bridge cells and matched-index and prospective family panels. These are

empirical permutation stress results. They support the bounded description **first-order organizer, not universal cutoff**.

6. Simulation evidence

6.1 Design, methods and Monte Carlo reporting

The roles are separated: regular-law Wald checks; nonregular-model mechanism interventions; and supporting permutation stress comparisons in S2. All use intact paired units. S3 specifies the exact generating distributions, seeds, RNG consumption, retained denominators and source files. All main profiles are uncapped. The default IF density estimates use a Gaussian kernel with $h = 1.06\hat{\sigma}n^{-1/5}$. Full fits use the public API’s median-start reweighting solver, at most 100 iterations and location-change tolerance $10^{-10} \max(1, \hat{s})$. S1.7 explains why this is not itself the theorem’s negligible score-residual condition.

For influence values $\hat{\ell}_i$, every main Wald standard error uses $\{\sum_i (\hat{\ell}_i - \bar{\ell})^2 / [n(n-1)]\}^{1/2}$. Intervals are ordinary untruncated two-sided 95% Wald intervals, without a Fisher transform. Rates use valid retained replications; no failed draw is replaced. Binomial MCSE is $\sqrt{\hat{p}(1-\hat{p})/R}$. Wilson intervals describe Monte Carlo uncertainty. Paired contrasts use the sample SD of within-replication differences divided by $\sqrt{R}$.

6.2 Regular IID null checks

The normal-margin shared-sign null is $X = S|U|, Y = S|V|$, with independent standard normal U,V and shared independent Rademacher S. The t5 null has independent standard t5 margins. The strong-skew null has independent $X = \exp(1.1U)$ and $Y \sim \text{Gamma}(0.7, \text{scale} = 1)$. All target $\rho_P = 0$.

Table 1. Regular-IID Wald checks. Six existing cells are shown, including the two smaller-size rows omitted from the condensed PDF. Each uses 1,000 replications and nominal .05; intervals are Monte Carlo Wilson intervals.

Law	n	Rejection	MCSE	95% Wilson
Normal/sign	160	0.0520	0.0070	[0.0399, 0.0676]
Normal/sign	640	0.0440	0.0065	[0.0329, 0.0586]
t5	320	0.0750	0.0083	[0.0603, 0.0930]
t5	640	0.0590	0.0075	[0.0460, 0.0754]
Strong skew	640	0.1220	0.0103	[0.1031, 0.1437]
Strong skew	2560	0.0820	0.0087	[0.0666, 0.1006]

At n=640, normal-margin shared-sign and independent t5 results are compatible with .05. Strong-skew rejection remains elevated at n=2560. Fixed regularity therefore does not imply fast finite-sample convergence. The null cells do not exercise a generally nonzero first-order reference contribution.

6.3 Nonzero-effect regular skew coverage

Let $(\log X, \log Y)^\top$ be centered bivariate normal with marginal SD .6 and correlation .4. Independent split quadrature gives $\rho_P = .19347511885804566$, raw MAD .387121537597268 and Huber reference 1.079808860961767. At each size 2,000 paired datasets compare the full estimator/IF, the same estimator with direct-only SE, and an oracle with population references and its direct IF. The root seed is 2026090701; S3.1 gives its replication mapping.

Table 2. Coverage of nominal 95% intervals. The Wilson interval and SE/SD column refer to the complete IF. Full bias, spread, widths and Monte Carlo uncertainty for all methods are in S3.4.

n	Full	95% Wilson	Direct	Oracle	SE/SD
160	0.8580	[0.8420, 0.8726]	0.8535	0.8485	0.8397
640	0.9115	[0.8983, 0.9232]	0.9110	0.9050	0.8999
2560	0.9270	[0.9148, 0.9376]	0.9265	0.9290	0.9664

All complete-coverage intervals exclude .95. At $n=2560$, .0595 and .0135 of the complete intervals lie wholly below and wholly above truth. Oracle undercoverage prevents attributing all deficiency to reference estimation. The nonzero reference contribution has coefficient about -.149814; its variance and covariance with the direct term nearly cancel in this law (full IF variance 2.930344, direct-only 2.935125). This does not justify general omission of nuisance terms. S3.6 reports the post hoc population-SE and root-replay diagnostics with their limitations.

6.4 Four-stage reference pathway

On the same shared-sign datasets, compare: population references fixed; population scales fixed and locations fitted; population medians fixed with MAD and locations fitted; and complete fitting. The design uses $\tau \in \{.10, .40\}$, $n \in \{80, 640\}$, 2,000 datasets per cell, root seed 2026090901: 8,000 datasets and 32,000 stage records. The internal protocol preceded outcomes; it was not a third-party preregistration.

Table 3. Severe shared-sign case, tau=.10. Stage-aware and direct-only columns are empirical null rejection rates; S3.2 defines each stage’s IF.

n	Stage	Mean rho	Stage-aware	Direct-only
80	Fixed ref.	0.0006	0.0720	0.0720
80	Fixed scale	0.3804	0.2235	0.6785
80	Fixed median	0.3804	0.2235	0.6785
80	Full	0.8207	0.7550	0.9035
640	Fixed ref.	-0.0006	0.0525	0.0525
640	Fixed scale	0.1117	0.0330	0.4875
640	Fixed median	0.1117	0.0330	0.4875
640	Full	0.6306	0.5510	0.8270

Table 4. References and secondary effect, tau=.10, n=640. C is averaged from each replication’s actual C. CV columns are averages of the two replication-specific empirical CVs, using divisor n. These averages must not be multiplied with mean rho to reconstruct mean C. Each row uses 2,000 paired replications. S3.5 gives Monte Carlo SEs and the replay provenance.

Stage*	RMSE X	RMSE Y	Corr(TX,TY)	Mean C	Mean CVx	Mean CVy
R	0.00000	0.00000	NA	0.999994	0.10011	0.10017
S	0.04013	0.04014	0.98990	1.001573	0.10735	0.10741
M	0.04013	0.04014	0.98990	1.001573	0.10735	0.10741
F	0.27897	0.27872	0.98671	1.079550	0.26555	0.26504

*R: fixed reference; S: fixed scale; M: fixed median; F: full fit.

Tables 3–4 show that correlated location errors already arise with fixed population scale and that full fitting adds substantial amplification. In the realized severe cells, stages 2–3 are fully linear and equal

the sample means. Conservative stage-aware rejection at $n=640$ does not imply negligible profile bias. At $n=80$ the paired fixed-scale-to-full increase in stage-aware rejection is .5315 (MCSE .0111609); mean ρ increases by .4402536. The additional amplification is much smaller at $\tau=.40$, whose complete results are retained in S3.3. That control still has $f(0)=0$ and is not regular-theorem validation.

There were no invalid stage returns. Two full-fit roots were flagged at $\tau=.10$, $n=640$, replications 70 and 1622. Bracketed recomputation changed ρ by less than 4.7×10^{-7} , relative SE by less than .0031%, and neither rejection decision. Original outcomes and flags are retained in S3.6; this sensitivity result is not a general solver guarantee.

7. Permutation scope

Re-pairing leaves both marginal samples, medians, MADs and Huber references unchanged. It changes joint moments and the studentized statistic. The exact fixed-orbit identity is $\hat{C}_\pi = 1 + \hat{\rho}_{P,\pi} CV_n(\hat{a}) CV_n(\hat{b})$. It gives matching unstudentized ordering under matching tail conventions, not equality of arbitrary studentized p-values. Exact randomization validity requires group invariance; the weak null alone does not provide it. Theorem 1 is not a conditional permutation CLT. S2 supplies actual identities, method definitions, figures and supporting results under this restricted interpretation.

8. Discussion

The contribution is complete inference and a worked analysis of finite-sample limits for generated robust-reference profiles. The regular theorem supplies a pointwise foundation. The shared-sign construction shows how linked fitted reference errors can create severe distortion; the stages distinguish a direct location-error path from additional full-fit amplification. The regular skew study shows a separate limit: even oracle references do not remove its undercoverage. Numerical derivative checks and regression tests establish neither statistical calibration nor independent acceptance of the proof.

Local conditioning explains part of a transition but must be interpreted together with score noise, target projection, profile variation, direct-reference covariance and nonlinear remainder. Neither a sample-size threshold nor an index cutoff is supplied. Classical alternatives answer different scientific questions; different null hypotheses cannot be ranked using rejection rates alone. A bounded reference score does not make an uncapped profile correlation globally robust.

A real application remains optional. It requires a defensible independent unit, scientific pairing, sampling population, and documented clustering, missingness, selection and preprocessing. No dataset has yet passed that design review. Shared edges, repeated measures or clusters require a different influence/resampling argument; a small number of buildings cannot be treated as an automatic cluster-CLT justification. No application is claimed here.

S1–S3 now provide the technical argument, bounded permutation evidence and reproducibility details used by the main text. Independent mathematical review and clean-environment reproduction remain outstanding. Final author decisions on placement, affiliations, journal, availability statements and accurate AI-assistance disclosure are recorded separately from the scientific claims. A universal cutoff, new resampling method or general degeneracy theorem is not a prerequisite for this paper’s bounded contribution.

Appendix A. Conditions and proof of Theorem 1

R1 (sampling). Intact pairs are IID from a fixed law P .

R2 (quantiles). Each median is unique, raw MAD is positive, and the marginal density is continuous and strictly positive in neighborhoods of $m, m - d, m + d$.

R3 (nondegeneracy). Each population Huber root is unique, its active probability A is positive, and there is no mass at T or its clipping boundaries. Both profile variances and V_ρ are positive.

R4 (general-effect moments). For some $\eta > 0$, $E\{a^{4+\eta} + b^{4+\eta} + (ab)^{2+\eta}\} < \infty$. These are sufficient, not minimal, conditions.

R5 (empirical estimation). Selected nuisance estimates are measurable and consistent with the stated midpoint convention. The Huber score residual is zero or $o_P(n^{-1/2})$. Density estimates are uniformly consistent on fixed neighborhoods of the required boundaries. S1.5 gives the default-KDE route and S1.7 the numerical implementation qualification.

Local class control (the former R6). S1.3 identifies the indicator, Huber-score, radius, squared-radius, paired-product and weighted-sign classes; gives polynomial covers and square-integrable envelopes under R4; and verifies their localized IF and squared-IF GC closures, including random fitted coefficients. This replaces a bare appeal to an unspecified R6 condition.

Proof. The paired Bahadur expansions and Huber Z-expansion in S1.1–S1.2 give the complete nuisance influences. S1.3 supplies random-index stochastic equicontinuity; S1.4 verifies the population Taylor remainders and hence the joint five-moment expansion. The ordinary correlation delta method and IID CLT give the first conclusion. S1.5 proves plug-in population L2 consistency and in-sample empirical variance consistency using the squared-class GC argument. Empirical centering and Slutsky give studentization. Every covariance is taken over intact pairs. This is a fixed-law proof, with no asserted uniform bound on nuisance conditioning or on the remainder. The reduced C and weak-null moment arguments are given separately in S1.6. $\square$

Appendix B. Proof of Proposition 1

Let G_τ be the CDF of $\exp(\tau U)$. For $x > 0$, $F_X(x) = 1/2 + G_\tau(x)/2 > 1/2$; for $x < 0$, $F_X(x) = \{1 - G_\tau(-x)\}/2 < 1/2$. Hence the unique median is zero and the median of the absolute radius is one. Symmetry and the odd Huber score give $T=0$. Every finite clipping interval has positive probability, so the population score is strictly decreasing and this root is unique. Independence of the two positive radii gives $\rho_P = 0, C = 1$.

For x nonzero, $f_X(x) = \phi\{\log|x|/\tau\}/(2\tau|x|)$, which tends to zero at zero; its values at the MAD boundaries are positive. On an unbalanced sign sample, the absolute midpoint median is at least the smallest observed radius. Thus for every fixed $M > 0$,

$$P\{\sqrt{n}|\hat{m}| \leq M\} \leq P(\text{exact sign balance}) + nG_\tau(M/\sqrt{n}) \longrightarrow 0.$$

The first term is $O(n^{-1/2})$ for even n and zero for odd n ; the Gaussian lower-tail decay in the second dominates n . Finally $A = \Phi\{\log(ck)/\tau\}$, giving score slope $-A/k$. Failure of median regularity identifies a failed sufficient condition of Theorem 1; it does not preclude an alternative asymptotic theory for a composite functional. Population root uniqueness is not a guarantee of uniqueness for every finite sample root. $\square$

Supplementary materials supplied with this draft

- [S1](#): paired proof, local classes, midpoint MAD, density plug-ins, weak-null qualification.
- [S2](#): permutation identities and scope, actual bridge and family-residual panels.
- [S3](#): generating laws, studentizers, source and seed ledgers, full main-study tables, diagnostics.

The materials are present in this reading package. Their eventual journal placement remains an author decision; legacy frozen display identifiers are preserved in the source ledger.

Supplement S1. Regular inference: a paired Bahadur and Z-estimation proof

This supplement supplies the proof details used by Appendix A of the integrated author draft. It concerns the uncapped profiles and a fixed regular IID law. It is an author-level mathematical argument, not independent mathematical peer review. The proof uses finite-dimensional quantile and Z-estimation expansions; it does not require empirical measures to converge in total variation, or assert a joint Hadamard theorem on an unspecified domain.

S1.1 Conditions and conventions

Use Conditions R1–R5 in Appendix A: intact IID pairs; unique median and positive raw MAD; continuous positive densities near the median and both MAD boundaries; a unique Huber root with positive active probability and no boundary atoms; positive profile variances and target IF variance; the stated moments; consistent measurable empirical roots and negligible Huber score residual; and locally uniformly consistent density estimates. All statements below are pointwise at that law. Local parameter sets are deterministic and compact, with scales and limiting denominators bounded away from zero.

For even sample size the median is the average of the two central order statistics. The MAD applies that same convention to deviations from the sample midpoint median. For odd size the middle order statistic is used. There is no finite-sample correction to the raw MAD; the fixed factor $k = 1.4826$ is applied afterward. This is the sample construction treated by Mazumder and Serfling (2009). The proof uses its weak Bahadur expansion, not a claim that sample quantiles solve indicator equations exactly.

Write P_n for the empirical law of intact pairs and $\mathbb{G}_n = \sqrt{n}(P_n - P)$. For a margin put $f_0 = f(m)$, $f_+ = f(m + d)$, and $f_- = f(m - d)$. The paired versions of the marginal Bahadur expansions are

$$\begin{aligned}\sqrt{n}(\hat{m} - m) &= \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{1/2 - 1\{W_i \leq m\}}{f_0} + o_P(1), \\ \sqrt{n}(\hat{d} - d) &= \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{1/2 - 1\{|W_i - m| \leq d\} - (f_+ - f_-)IF_m(W_i)}{f_+ + f_-} + o_P(1).\end{aligned}$$

Each marginal remainder is negligible jointly because there are only two margins. Their leading sums are paired and may be correlated. Positive continuous boundary densities supply the local quantile identification needed for these expansions. The midpoint convention is not silently replaced by a lower empirical quantile.

S1.2 Huber equation and the paired score system

Use the following fixed estimating-equation normalization, with $\theta_W = (m_W, d_W, T_W)^\top$:

$$\xi_W(w; \theta_W) = \begin{pmatrix} 1\{w \leq m_W\} - 1/2 \\ 1\{|w - m_W| \leq d_W\} - 1/2 \\ \psi_c\{(w - T_W)/(kd_W)\} \end{pmatrix}.$$

Let $U = (W - T)/(kd)$, $A = P(|U| < c)$, and $B = P\{U1(|U| < c)\}$. The population derivative in the unscaled coordinates is

$$J_{0W} = \begin{pmatrix} f_0 & 0 & 0 \\ f_+ - f_- & f_+ + f_- & 0 \\ 0 & -B/d & -A/(kd) \end{pmatrix}.$$

The Huber equation is differentiable in (T, s) at the population values: $\partial_T P\psi_c = -A/s$ and $\partial_s P\psi_c = -B/s$. The score is bounded and locally Lipschitz in location and positive scale. Boundary no-mass conditions justify the displayed derivatives. Consistency, local stochastic equicontinuity (S1.3), the MAD expansion, and $P_n\psi_c\{(W - \hat{T})/(k\hat{d})\} = o_P(n^{-1/2})$ first give the root-n rate from the nonzero location derivative, then

$$\sqrt{n}(\hat{T} - T) = \frac{1}{\sqrt{n}} \sum_i \left[\frac{s}{A} \psi_c\{(W_i - T)/s\} - \frac{B}{A} kIF_d(W_i) \right] + o_P(1).$$

Define $\xi_{XY}(Z) = (\xi_X(X)^\top, \xi_Y(Y)^\top)^\top$ and $\Omega_{XY} = E(\xi_{XY}\xi_{XY}^\top)$. The mean score is zero. The derivative is $J_{0,XY} = \text{diag}(J_{0X}, J_{0Y})$, but Ω_{XY} is generally not block diagonal. Thus

$$\sqrt{n}(\hat{\theta}_{XY} - \theta_{XY}) = -J_{0,XY}^{-1} \frac{1}{\sqrt{n}} \sum_i \xi_{XY}(Z_i) + o_P(1).$$

For a margin, standardize the *error* as $e_W = (\hat{\theta}_W - \theta_W)/d_W$, using the population raw MAD. Then $J_W = d_W J_{0W}$ and $\text{Var}(e_W) \approx J_W^{-1} \Omega_W J_W^{-\top} / n$, where $\Omega_W = E(\xi_W \xi_W^\top)$. This standardizes a local perturbation; it is not the derivative of the nonlinear coordinate map $(m/d, d/d, T/d)$. Arbitrary rescaling of the score equations changes the singular values of J_W . Positive affine changes of data units leave the declared standardized derivative unchanged. Target projection and paired score covariance remain necessary for the target variance.

S1.3 Verification of the local empirical-process classes

Here is the class verification used for R6, under R1–R5. Use a compact neighborhood with $|T_X|, |T_Y| \leq M$ and $d \geq d_0 > 0$. Measurable versions (or the usual separable versions of these finite-parameter classes) are understood. Norms in this section are population norms unless P_n is written explicitly.

Indicators and signs. Half-lines and intervals are VC set classes. The classes $1\{w \leq t\}$, $1\{|w - m| \leq d\}$, and $1\{|w - T| < ckd\}$ therefore have polynomial covering numbers with envelope one. Signs are affine transformations of half-line indicators, with the value at equality treated separately on a probability-zero boundary.

Clipped and active scores. The clipped Huber score is bounded by c . On $d \geq d_0$, its location derivative is bounded and its scale derivative is bounded because an active standardized residual has magnitude at most c . It is a finite-dimensional Lipschitz class with a constant envelope. The active score $u1\{|u| < c\}$ is bounded by c but jumps at its knots. On each of the finitely many regions separated by $T \pm ckd$, its subgraph inequalities become linear inequalities after multiplication by the positive kd . Their finite unions and intersections have finite VC dimension. No mass at the limiting knots gives $L_2(P)$ continuity.

Translated radii and moments. The following pointwise bounds establish Euclidean parameter covers with the indicated envelopes:

$$\begin{aligned} ||x - t| - |x - t'|| &\leq |t - t'|, \\ |(x - t)^2 - (x - t')^2| &\leq |t - t'|(2|x| + 2M), \\ ||x - t||y - u| - |x - t'||y - u'| &\leq |t - t'|(|y| + M) + |u - u'|(|x| + M). \end{aligned}$$

Envelopes are $|X| + M$, $(|X| + M)^2$, and $(|X| + M)(|Y| + M)$, respectively. The general-correlation moments in R4 make these square integrable. For the reduced three-moment vector used for C , omit the squared-radius classes; the reduced moments in S1.6 suffice.

Weighted moving signs. For example, $\text{sign}(x - t)|y - u|$ has envelope $|Y| + M$. Partition its subgraph by $x \leq t$ and $y \leq u$. On each of four regions the function equals one of $y - u$, $u - y$, $-(y - u)$, $-(u - y)$. The subgraph is a finite Boolean combination of half-spaces in (x, y, z) ; consequently the class has polynomial VC covers. Its continuity at (T_X, T_Y) follows from dominated convergence: the squared difference is bounded by a constant times $1 + Y^2$, and pointwise convergence fails only at the probability-zero moving-sign boundary. The other margin is identical with X and Y interchanged. Dependence within a pair does not invalidate domination.

Finite IF closure and random coefficients. Treat every fitted scalar coefficient (density evaluations, A , B , moments, and smooth ratios) as an additional coordinate in a deterministic compact set containing its limit. Denominators are bounded away from zero on that set. Finite sums of the classes above and multiplication by bounded scalar coefficients retain polynomial entropy: cover each summand at accuracy proportional to the desired total accuracy and take the product of the finitely many covers. This argument does not assert that arbitrary products of unbounded Donsker classes are Donsker.

For the full IF an envelope is $H = K(1 + |X| + |Y| + X^2 + Y^2 + |XY|)$; for the reduced IF omit X -squared and Y -squared. The appropriate moment condition gives $PH^2 < \infty$. Polynomial entropy and the square-integrable envelope give the needed Donsker conclusion. They also give Glivenko–Cantelli control. To handle the squared IF class, truncate all influences to $[-L, L]$, use the Lipschitz square map there, and bound the discarded tails uniformly by $H^2 1\{H > L\}$. Its population expectation tends to zero; for each fixed L its empirical mean converges by the ordinary law of large numbers. Letting L grow proves the squared-class GC assertion with an integrable envelope, without requiring an eighth moment.

These are the VC/Euclidean entropy and closure routes of van der Vaart and Wellner (1996) and van der Vaart (1998). After localization and the displayed $L_2(P)$ continuity, estimated-function stochastic equicontinuity gives $\mathbb{G}_n(f_{\hat{\theta}} - f_{\theta_0}) = o_P(1)$ (van der Vaart and Wellner, 2007). Thus R6 is connected to concrete classes and envelopes rather than left as an unexplained assumption. Independent review of this verification remains outstanding.

S1.4 Joint moments and the population Taylor remainder

Put $a = |X - T_X|$, $b = |Y - T_Y|$. Away from an atom at the reference, the population derivative of $P|X - t|$ at T_X is $-P \operatorname{sign}(X - T_X)$. One useful remainder bound is

$$||x - T - h| - |x - T| + h \operatorname{sign}(x - T)| \leq 2|h|1\{|x - T| \leq |h|\}.$$

Its expectation is $o(|h|)$. In the paired product the corresponding bound is weighted by b (or a); integrability and no boundary atom imply $P[b1\{|X - T_X| \leq |h|\}] \rightarrow 0$. The product of the two residual changes is bounded by $|h_X h_Y|$. The squared-radius expansion is algebraically exact up to the shift squared. With root- n nuisance errors these give $o_P(n^{-1/2})$ population remainders. S1.3 controls the empirical random-index remainders. Therefore

$$\sqrt{n}(\widehat{M} - M) = n^{-1/2} \sum_i IF_M(Z_i) + o_P(1), \quad M = (\nu, \mu_a, \mu_b, q_a, q_b)^\top,$$

with the complete influences in Section 3. The finite-dimensional correlation map is differentiable at positive profile variances. Its gradient gives $IF_\rho = \nabla \rho_P^\top IF_M$; R4 makes its variance finite. The IID CLT then proves the unstudentized conclusion of Theorem 1.

S1.5 Density plug-ins and empirical variance consistency

Uniform density consistency near $m, m \pm d$, combined with consistent random evaluation points and continuity of f , yields consistency of every plug-in boundary density. For the implemented Gaussian KDE,

$$\widehat{f}_h(t) = \frac{1}{nh} \sum_i \phi\{(W_i - t)/h\}, \quad \widehat{h} = 1.06\widehat{\sigma}_W n^{-1/5},$$

$\widehat{\sigma}_W$ is the ordinary sample standard deviation with divisor $n - 1$. Finite nonzero variance gives $\widehat{h}/(1.06\sigma_W n^{-1/5}) \rightarrow_P 1$. On a probability-tending-to-one event h lies in a deterministic band $[c_1 n^{-1/5}, c_2 n^{-1/5}]$. Gaussian translation/scale entropy yields a uniform stochastic bound over that band tending to zero, since $nh/\log n \rightarrow \infty$. Local uniform continuity controls kernel bias on a slightly smaller compact neighborhood; the Gaussian tails and integrability of the density control the outside contribution. This supplies a route to R5 for the default KDE, with the uniform kernel-density theory of Giné and Guillou (2002). Consistency at random boundaries is not deduced solely from fixed-point KDE consistency. Analytic and cross-fit options exist in the software; they are not the default method in the main studies and require their own matching conditions if used.

Localize the fitted scalar coefficients as in S1.3. Pointwise convergence outside null boundaries and domination by H give $\|\widehat{IF}_\rho - IF_\rho\|_{P,2} = o_P(1)$. For the in-sample variance,

$$\begin{aligned} P_n \widehat{IF}_\rho^2 - P IF_\rho^2 &= (P_n - P)(\widehat{IF}_\rho^2 - IF_\rho^2) \\ &\quad + P(\widehat{IF}_\rho^2 - IF_\rho^2) + (P_n - P)IF_\rho^2. \end{aligned}$$

The terms vanish by squared-class GC, Cauchy–Schwarz and function-norm consistency, and the ordinary LLN, respectively. The first-moment argument justifies empirical centering. Thus the sample variance of the paired plug-in influences is consistent; Slutsky yields the Wald conclusion.

S1.6 Secondary C and the weaker-moment null route

For C use only $M_C = (\nu, \mu_a, \mu_b)^\top$ and $E\{a^{2+\eta} + b^{2+\eta} + (ab)^{2+\eta}\} < \infty$ for some $\eta > 0$. The reduced class, Taylor expansion and variance argument give

$$IF_C = \frac{IF_\nu}{\mu_a \mu_b} - C \frac{IF_{\mu_a}}{\mu_a} - C \frac{IF_{\mu_b}}{\mu_b}.$$

No root-n CLT for q_a, q_b is imported under these reduced conditions. At $\rho_P = 0$, the correlation numerator uses the same three-moment expansion and its denominator needs only consistency. The squared-moment gradient coefficients vanish at the null, giving the reduced null influence.

For the unrestricted empirical studentizer, $\hat{\rho}_P = O_P(n^{-1/2})$. Finite second moments imply

$$\frac{P_n a^4}{n} \leq \frac{\max_i a_i^2}{n} P_n a^2 = o_P(1), \quad \hat{\rho}_P^2 P_n a^4 = o_P(1).$$

For fitted radii use $\hat{a}_i \leq a_i + |\hat{T}_X - T_X|$, hence $P_n \hat{a}^4 \leq 8P_n a^4 + 8|\hat{T}_X - T_X|^4$; the same control applies. Empirical Cauchy–Schwarz controls cross terms with the reduced IF. This establishes empirical-norm negligibility of the extra quadratic terms. When fourth moments are infinite it does **not** establish population $L_2(P)$ consistency of an unrestricted plug-in IF having a nonzero squared-profile coefficient. This null argument is not a general nonzero effect confidence theorem.

S1.7 Software boundary

The mathematical procedure requires exact or sufficiently accurate empirical Huber roots. The public API implements iteratively reweighted location updates from the sample median, at most 100 iterations, stopping when the location change is below $10^{-10} \max(1, \hat{s})$. It does not enforce an n-dependent score-residual bound. Neither a fixed tolerance nor these finite replays prove the asymptotic root condition. Section S3.6 records the two flagged full fits and their bracketed sensitivity checks. The public solver and its defaults have not been changed during manuscript integration.

This closes the presentation of the proof route, local classes, moving boundaries, random coefficients, and density evaluation. It does not certify independent mathematical acceptance or a uniform software convergence theorem.

Supplement S2. Permutation scope and empirical conditioning evidence

This supplement supplies the evidence cited in Sections 5 and 7. All permutation rejection frequencies below are **empirical null rejection rates**, not a demonstrated weak-null Type I error guarantee. The figures retain their legacy Figure 2 and Figure 3 identifiers; including them here does not record supervisor approval of their eventual journal placement.

S2.1 Exact fixed-margin identities

For fixed nonconstant nonnegative profile vectors a, b with positive means, use empirical variances with divisor n . Permuting b preserves both means, variances and CVs. Expanding the empirical covariance gives

$$\hat{C}_\pi = \frac{\bar{a}\bar{b} + \text{Cov}_n(a, b_\pi)}{\bar{a}\bar{b}} = 1 + \hat{\rho}_{P,\pi} CV_n(a) CV_n(b).$$

The multiplier is positive. Thus one-sided ranks agree, as do two-sided ranks based on absolute distance from the respective null values 1 and 0. This is an unstudentized identity, not an equivalence for arbitrary studentizers or tail definitions. Re-pairing raw X and Y also leaves each marginal sample unchanged, so the median, MAD and Huber fit are unchanged. The Y -indexed marginal IF vector is permuted along with Y ; joint moments, target reference coefficients and empirical IF variance are recomputed.

Under a permutation-invariant null law, the rank of the observed statistic on its orbit supplies a valid randomization test with the declared tie convention. IID pairs with independent X and Y supply such pairing invariance. The larger null $\rho_P = 0$ does not. In particular, shared signs can make raw variables dependent despite independent population radii. This manuscript does not prove a conditional permutation CLT for its weak null. The observed Wald CLT in Theorem 1 is a different statement. Multiplicity adjustment cannot repair an invalid local p-value; Holm does not require subset pivotality.

S2.2 Empirical permutation method

The frozen method uses complete five-moment profile influences with Gaussian KDE boundary estimates and the same median/MAD/Huber constants as the main text. For each of B sampled pairings it recomputes joint moments and the complete studentized statistic Z_π . It reports

$$p_{MC} = \frac{1 + \sum_{j=1}^B 1\{|Z_{\pi_j}| \geq |Z_{obs}|\}}{B + 1},$$

and rejects when $p_{MC} \leq .05$. Here $B=99$. This formula describes the calculation, not a proof of weak-null calibration. Wilson intervals use the number of independent generated datasets, not B as the denominator. The function is `profile_studentized_permutation_test` in `scripts/run_studentized_permutation_weak_null_20260814.py`.

S2.3 Bridge generating laws and evidence

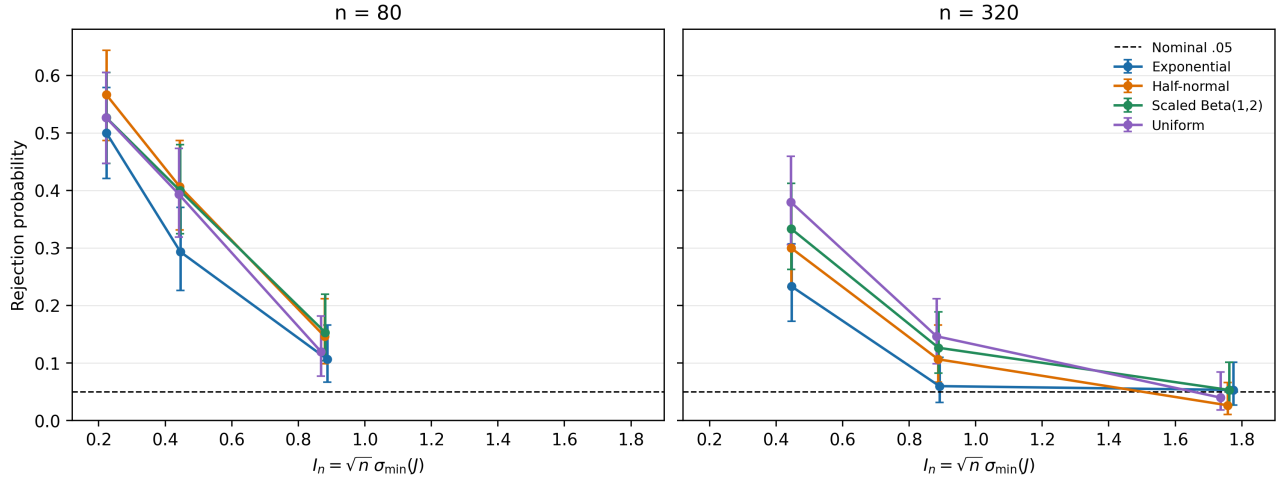
Let $X = SR_X, Y = SR_Y$. S is a symmetric sign independent of two independent radii. Each radius is independently drawn from

$$(1 - \epsilon)\mathcal{L}\{\exp(.10U)\} + \epsilon Q_f, \quad U \sim N(0, 1).$$

The bridge distributions Q_f are Uniform(0,1), Exponential(rate=1), $|N(0, 2/\pi)|$ (normal variance $2/\pi$), and 2Beta(1, 2). All have right density one at zero, giving symmetric marginal $f(0) = \epsilon/2$. Their independent radii give population $T_X = T_Y = 0$, $\rho_P = 0, C = 1$. Their MADs and Huber active fractions are evaluated for each mixture; the standardized Jacobian is not replaced by the median density alone.

The 24-cell panel uses four families, $n=80, 320$ and $\epsilon=.05, .10, .20$. Each cell has 150 datasets, 99 permutations and 199 bootstrap reference fits per margin for historical diagnostics. The bootstrap fits are not used to define the displayed population index. Their RNG consumption must nevertheless be retained to reproduce the original cell stream. S3.7 gives exact cell seeds and implementation sources.

Figure 2. Frozen bridge transition: fully recomputed studentized-permutation evidence only (150 replications and 99 permutations per cell; empirical, not Wald-theorem validation)



Legacy Figure 2. Bridge transition: empirical studentized-permutation evidence only. Whiskers are Wilson intervals across datasets.

Table S2.1. Full bridge panel. Index is population I_n . Rates and intervals describe empirical permutation rejection. These rows are the actual data behind legacy Figure 2, not a proposed future display.

Family	n	epsilon	Index	Rejection	95% Wilson
Exp	80	0.05	0.2232	0.5000	[0.4210, 0.5790]
Half-normal	80	0.05	0.2228	0.5667	[0.4867, 0.6433]
Beta	80	0.05	0.2229	0.5267	[0.4471, 0.6049]
Uniform	80	0.05	0.2222	0.5267	[0.4471, 0.6049]
Exp	80	0.1	0.4456	0.2933	[0.2264, 0.3706]
Half-normal	80	0.1	0.4437	0.4067	[0.3313, 0.4867]
Beta	80	0.1	0.4442	0.4000	[0.3250, 0.4800]
Uniform	80	0.1	0.4412	0.3933	[0.3188, 0.4732]
Exp	80	0.2	0.8872	0.1067	[0.0667, 0.1662]
Half-normal	80	0.2	0.8787	0.1467	[0.0989, 0.2121]
Beta	80	0.2	0.8809	0.1533	[0.1044, 0.2196]
Uniform	80	0.2	0.8681	0.1200	[0.0773, 0.1817]
Exp	320	0.05	0.4464	0.2333	[0.1728, 0.3072]
Half-normal	320	0.05	0.4455	0.3000	[0.2324, 0.3776]
Beta	320	0.05	0.4458	0.3333	[0.2629, 0.4121]
Uniform	320	0.05	0.4443	0.3800	[0.3062, 0.4598]
Exp	320	0.1	0.8912	0.0600	[0.0319, 0.1101]
Half-normal	320	0.1	0.8873	0.1067	[0.0667, 0.1662]
Beta	320	0.1	0.8883	0.1267	[0.0826, 0.1894]
Uniform	320	0.1	0.8824	0.1467	[0.0989, 0.2121]
Exp	320	0.2	1.7744	0.0533	[0.0273, 0.1017]
Half-normal	320	0.2	1.7574	0.0267	[0.0104, 0.0666]
Beta	320	0.2	1.7618	0.0533	[0.0273, 0.1017]
Uniform	320	0.2	1.7361	0.0400	[0.0185, 0.0845]

Across the displayed bridge cells rejection generally declines as the population index increases. This is an empirical ordering, not direct Wald theorem validation, a universal cutoff, or a reliability certificate.

S2.4 Matched-index and prospective family residuals

The matched-index confirmation uses $n=320$, $\epsilon=0.05$, 500 datasets per family and 99 permutations. The family comparison was selected after the pilot and uses a larger replication run. In the original implementation the pilot and confirmation share a cell seed; their first 150 datasets overlap. Accordingly this is a larger-run confirmation, not an entirely independent replication of the pilot. It remains separate from the September 9 Wald pathway study. The source scripts and seed ledger make that overlap explicit.

Table S2.2. Matched-index confirmation.

Family	Index	Rejection	95% Wilson
Exp	0.4464	0.2320	[0.1971, 0.2710]
Half-normal	0.4455	0.3100	[0.2710, 0.3519]
Beta	0.4458	0.3560	[0.3153, 0.3989]
Uniform	0.4443	0.3340	[0.2941, 0.3765]

The scalar index does not determine the rejection rate. Even exact equality of J would not equate score covariance, target projection, profile variability, or direct-reference covariance; differences are

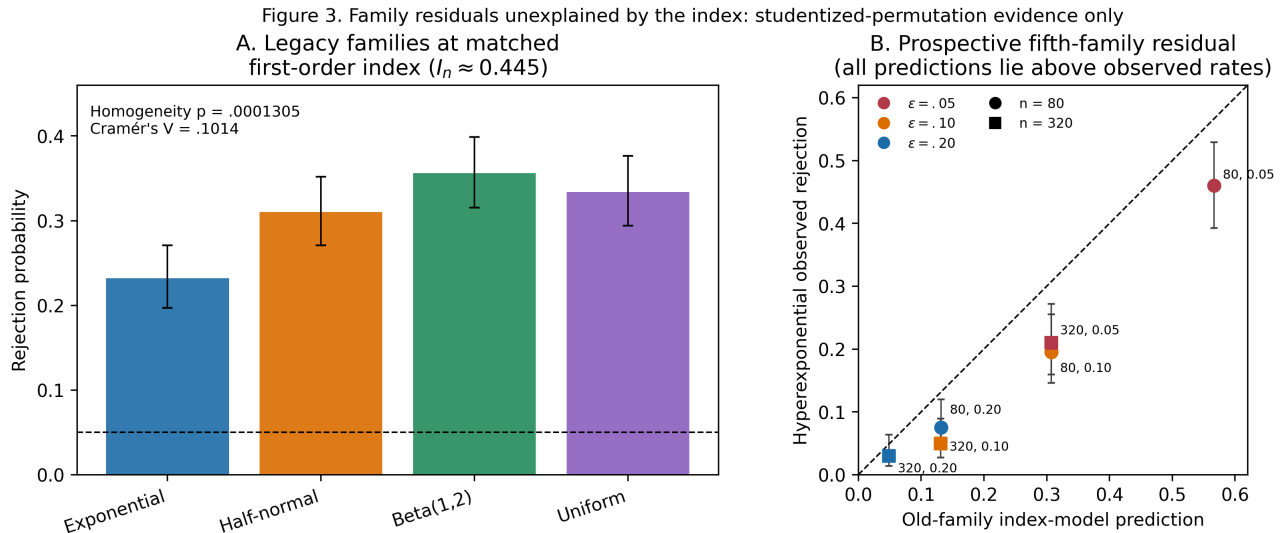
not identified as purely higher order.

The prospective fifth bridge is the mixture $Q_H = .5 \text{Exp}(.5) + .5 \text{Exp}(1.5)$, with rates in parentheses. Its right density at zero is also one. The predictor is a binomial-logit model with intercept and log population index, fitted on the old bridge panel before simulating the fifth family's outcomes. It uses six cells ($n=80,320$; $\epsilon=.05,.10,.20$), each with 200 datasets and 99 permutations. No bootstrap-reference diagnostic is consumed in this fifth family's stream.

Table S2.3. Prospective predictions and observed rejection.

n	epsilon	Index	Predicted	Observed	95% Wilson
80	0.05	0.2234	0.5663	0.4600	[0.3923, 0.5292]
80	0.1	0.4462	0.3074	0.1950	[0.1461, 0.2554]
80	0.2	0.8898	0.1314	0.0750	[0.0460, 0.1200]
320	0.05	0.4467	0.3070	0.2100	[0.1593, 0.2716]
320	0.1	0.8923	0.1309	0.0500	[0.0274, 0.0896]
320	0.2	1.7795	0.0489	0.0300	[0.0138, 0.0639]

All six predictions exceed the observations. This supports transport of coarse ordering while preserving a family residual; it does not calibrate the levels or produce an operational rule.



Legacy Figure 3. Family residuals unexplained by the index. Both panels use empirical studentized-permutation evidence.

Figure 3's homogeneity p-value and Cramér's V describe the legacy family comparison. They are not weak-null validity results for profile correlation. The accompanying table sources are the frozen September 5 display tables; their underlying source hashes are verified by the existing display audit. No family grid has been expanded in assembling this supplement.

Supplement S3. Designs, studentizers, numerical diagnostics and sources

This supplement supplies the reproducibility material behind Section 6. No fresh Monte Carlo cells were generated for this integration. A bounded deterministic replay of 2,000 existing September 9 datasets is used only to recover marginal profile CVs at stored references; S3.5 reports its checks. Numerical summaries below are generated from existing result files, rather than manually transcribed into independent tables.

S3.1 Main generating laws and random-number mapping

All main studies use independent units, one intact X/Y pair per unit. Within-pair dependence is specified explicitly. The population effects are the uncapped robust-reference effects in Section 2.

1. **Normal-margin shared-sign null:** independently generate a Rademacher sign S and U, V standard normal; set $X=S|U|$ and $Y=S|V|$. Both margins are standard normal; the raw pair is dependent and population radii are independent. $n=160,640, 1,000$ datasets each.
2. **Independent t5 null:** X, Y are independent standard t distributions with five degrees of freedom. $n=320,640, 1,000$ datasets each.
3. **Independent strong-skew null:** $X=\exp(1.1U)$, U standard normal, and independent Y is $\text{Gamma}(\text{shape}=.7, \text{scale}=1)$. $n=640,2560, 1,000$ datasets each. These first three laws have $\rho=0$ and use the complete IF Wald test.
4. **Nonzero correlated lognormal:** independently across units generate $(\log X, \log Y)$ centered bivariate normal with marginal SD .6 and correlation .4. $n=160,640, 2,000$ paired datasets each, root seed 2026090701. Each replication uses PCG64 with `SeedSequence([root,n,rep])`, $\text{rep}=0, \dots, 1999$. Full, direct-only and population-reference oracle methods share the draw.
5. **Four-stage shared-sign:** $X=S \exp(\tau U)$, $Y=S \exp(\tau V)$, with independent standard U, V and independent common symmetric S . $\tau=.10, .40$ and $n=80,640, 2,000$ datasets per cell. PCG64 uses `SeedSequence([2026090901, round(100*\tau), n, rep])`. Within a draw the sign vector is generated first, then the X normal vector, then the Y normal vector. All four stages use the same pair array.

Table S3.1. Exact cell seeds for the six regular-null cells. The original run uses one sequential NumPy `default_rng/PCG64` stream per cell, not the September 7 replication-level `SeedSequence` mapping. The seed is `2026082301+100000*j+n`, where $j=1, \dots, 6$ is the row order below.

j	Law	n	Cell seed
1	Normal/sign	160	2026182461
2	Normal/sign	640	2026282941
3	t5	320	2026382621
4	t5	640	2026482941
5	Strong skew	640	2026582941
6	Strong skew	2560	2026684861

The reference generator is `generate_regular_scenario` in `scripts/audit_wald_convergence_20260822.py`; the six-cell selection is `run_claim1` in `scripts/run_claim_validation_20260823.py`. The nonzero study uses `simulate` in `scripts/validate_active_nuisance_20260907.py`. It draws an n -by-2 standard normal array z , then sets $X=\exp(.6*z[:,0])$ and $Y=\exp(.6*(.4*z[:,0]+\sqrt{1-}$

.4**2)*z[:,1])). Replacing this with a different bivariate-normal sampler need not preserve the frozen stream. The four-stage generator is `generate` in `scripts/run_reference_pathway_20260909.py`.

The September 7 manifest records Python 3.12.3, NumPy 1.26.4 and SciPy 1.16.3. These are provenance entries, not a claim that a clean environment with those versions has been exercised here. A seed alone is insufficient to reproduce a different RNG call order or changed numerical libraries.

S3.2 Exact studentizers and numerical methods

For arbitrary reference values t_X, t_Y , define centered profiles $a_c = a - \text{mean}(a)$, $b_c = b - \text{mean}(b)$, divisor- n variances v_a, v_b and $D = \sqrt{v_a v_b}$. The direct correlation IF is

$$\ell_{0i} = \frac{a_{ci}b_{ci}}{D} - \frac{\hat{\rho}_P}{2} \left(\frac{a_{ci}^2}{v_a} + \frac{b_{ci}^2}{v_b} \right).$$

With fitted sample $g_X = \text{mean}(\text{sign}(X - t_X))$ and $h_X = \text{mean}(\text{sign}(X - t_X)b)$, put

$$\hat{\lambda}_X = \frac{-\hat{h}_X + \hat{g}_X \bar{b}}{D} + \frac{\hat{\rho}_P}{v_a} \{P_n(X - t_X) - \bar{a} \hat{g}_X\},$$

and define $\lambda_{\lambda Y}$ by interchanging margins. The stage-aware IF is $\ell_i = \ell_{0i} + \hat{\lambda}_X \hat{IF}_{T_X, i} + \hat{\lambda}_Y \hat{IF}_{T_Y, i}$. The two studentizers are sample $\text{SD}(\ell)/\sqrt{n}$ and sample $\text{SD}(\ell_{0i})/\sqrt{n}$, both with variance divisor $n-1$. The stage-aware label describes the formula, not theorem validity for the nonregular full fit.

- **Stage 1, population reference:** $t_X = t_Y = 0$; location influences are zero. The scale k is recorded descriptively; no Huber score equation is fitted.
- **Stage 2, population scale:** $s = k$ fixed. Fit the Huber location and use $IF_T = (s/A)\psi_c$. Plug in the empirical active fraction A .
- **Stage 3, population median:** fix $m = 0$ and estimate $d = \text{median}(\text{abs}(W))$; fit the location at $s = kd$. Use $IF_d = \{.5 - 1(|W| \leq d)\} / \{f(d) + f(-d)\}$, then $IF_T = (s/A)\psi_c - (B/A)kIF_d$. The absent median term is not estimated.
- **Stage 4, full fit:** use the complete median/MAD/Huber IF in Section 3 with all empirical joint moments and coefficients.

Stages 2–3 use SciPy `brentq` on $[\min(W), \max(W)]$ with `xtol=1e-13`. Stage 4 uses the unchanged public API. The default Gaussian KDE uses the sample SD bandwidth in S1.5 and evaluations at the fitted median and its two MAD boundaries (stage 3: at plus/minus d). The full solver starts at the midpoint median and performs at most 100 reweighted updates. The default c and k are 1.345 and 1.4826. Zero MAD, a degenerate active fraction or profile variance, or invalid SE is not interpreted as a valid zero effect. The saved error fields retain invalid returns; no replacements occurred.

Tests use two-sided $|\rho/\text{SE}| > 1.959963984540054$ in the pathway study; the regular API uses its corresponding normal p -value and nominal .05. The nonzero study covers truth when the ordinary untruncated 95% interval includes .19347511885804566. Direct-only methods omit reference IF terms but retain the fitted estimator; the oracle uses population locations and the direct IF matching those fixed profiles. No permutation, bootstrap correction, Fisher transform or tuned coverage adjustment is used in the main studies.

S3.3 Full four-stage results

Every row has 2,000 valid records and zero failures. R denotes population reference, S population scale, M population median, and F full fitting. Both tau values remain nonregular at the median.

Table S3.2. Profile effects and stage-aware rejection. The interval is the 95% Wilson interval for stage-aware empirical null rejection.

tau	n	Stage	Mean rho	Rejection	95% Wilson
0.1	80	R	0.0006	0.0720	[0.0615, 0.0842]
0.1	80	S	0.3804	0.2235	[0.2058, 0.2423]
0.1	80	M	0.3804	0.2235	[0.2058, 0.2423]
0.1	80	F	0.8207	0.7550	[0.7357, 0.7733]
0.1	640	R	-0.0006	0.0525	[0.0436, 0.0632]
0.1	640	S	0.1117	0.0330	[0.0260, 0.0418]
0.1	640	M	0.1117	0.0330	[0.0260, 0.0418]
0.1	640	F	0.6306	0.5510	[0.5291, 0.5727]
0.4	80	R	-0.0035	0.0755	[0.0647, 0.0879]
0.4	80	S	0.0616	0.0490	[0.0404, 0.0594]
0.4	80	M	0.0616	0.0485	[0.0399, 0.0588]
0.4	80	F	0.0693	0.0430	[0.0350, 0.0528]
0.4	640	R	0.0007	0.0615	[0.0518, 0.0729]
0.4	640	S	0.0102	0.0505	[0.0417, 0.0610]
0.4	640	M	0.0102	0.0505	[0.0417, 0.0610]
0.4	640	F	0.0101	0.0505	[0.0417, 0.0610]

Table S3.3. Direct-only rejection and reference errors. Reference correlation is across replications. Constant-reference correlation is NA.

tau	n	Stage	Direct rate	RMSE X	RMSE Y	Corr(TX,TY)
0.1	80	R	0.0720	0.00000	0.00000	NA
0.1	80	S	0.6785	0.11434	0.11481	0.99030
0.1	80	M	0.6785	0.11434	0.11481	0.99030
0.1	80	F	0.9035	0.50555	0.50593	0.98468
0.1	640	R	0.0525	0.00000	0.00000	NA
0.1	640	S	0.4875	0.04013	0.04014	0.98990
0.1	640	M	0.4875	0.04013	0.04014	0.98990
0.1	640	F	0.8270	0.27897	0.27872	0.98671
0.4	80	R	0.0755	0.00000	0.00000	NA
0.4	80	S	0.1605	0.13323	0.13266	0.86889
0.4	80	M	0.1610	0.13317	0.13248	0.86926
0.4	80	F	0.1780	0.14508	0.14539	0.89306
0.4	640	R	0.0615	0.00000	0.00000	NA
0.4	640	S	0.0785	0.04785	0.04763	0.87410
0.4	640	M	0.0785	0.04784	0.04762	0.87384
0.4	640	F	0.0770	0.04766	0.04736	0.86932

Table S3.4. Paired stage contrasts. Difference means the later stage minus the earlier stage, using the same 2,000 datasets. Entries show difference (paired MCSE); rejection is stage-aware. Direct-only and reference MSE contrasts remain in the source ledger for exact machine-readable use.

tau	n	Contrast	Delta rejection (MCSE)	Delta rho (MCSE)
0.1	80	R to S	0.1515 (0.01050)	0.3798 (0.00639)
0.1	80	S to M	0.0000 (0.00000)	-0.0000 (0.00000)
0.1	80	M to F	0.5315 (0.01116)	0.4403 (0.00574)
0.1	80	S to F	0.5315 (0.01116)	0.4403 (0.00574)
0.1	640	R to S	-0.0195 (0.00621)	0.1123 (0.00277)
0.1	640	S to M	0.0000 (0.00000)	0.0000 (0.00000)
0.1	640	M to F	0.5180 (0.01122)	0.5189 (0.00658)
0.1	640	S to F	0.5180 (0.01122)	0.5189 (0.00658)
0.4	80	R to S	-0.0265 (0.00574)	0.0651 (0.00177)
0.4	80	S to M	-0.0005 (0.00112)	-0.0001 (0.00004)
0.4	80	M to F	-0.0055 (0.00165)	0.0078 (0.00053)
0.4	80	S to F	-0.0060 (0.00187)	0.0077 (0.00053)
0.4	640	R to S	-0.0110 (0.00338)	0.0095 (0.00031)
0.4	640	S to M	0.0000 (0.00000)	-0.0000 (0.00000)
0.4	640	M to F	0.0000 (0.00000)	-0.0001 (0.00001)
0.4	640	S to F	0.0000 (0.00000)	-0.0001 (0.00001)

The original protocol’s statement of almost-sure empirical-root uniqueness was corrected after the run: a separated, exactly balanced sample can have an interval of Huber roots. Population uniqueness remains valid. The correction and the original outcomes are both retained. No invalid return occurred in these runs; that does not establish all-sample uniqueness.

S3.4 Nonzero-effect results and numerical truth

The population target is from split adaptive quadrature and analytic truncated-lognormal moments, not the earlier coarse .193218 benchmark. The variance quadrature splits at the nuisance knots and was checked at 24,48,72 nodes per interval. At the highest order the full variance is 2.930344036191777, direct-only variance 2.935125447074918, nuisance variance .029924671347371, and twice the direct–nuisance covariance -.034706082230512. The last two explain the small net variance difference without justifying omission of nuisance terms. Four contamination directions at epsilon=1e-6 give maximum scaled derivative discrepancy about 1.596e-5.

Table S3.5. Full nonzero-study summaries. F=complete IF, D=direct-only ablation, O=population-reference oracle. Each row has 2,000 valid estimates and zero failures. Bias is estimate minus population truth. SE is mean estimated standard error; SD is empirical sampling SD.

n	Method	Bias	Bias MCSE	SD	Mean SE	SE/SD
160	F	-0.00782	0.00278	0.12425	0.10434	0.8397
160	D	-0.00782	0.00278	0.12425	0.10342	0.8323
160	O	-0.00918	0.00280	0.12502	0.10336	0.8268
640	F	-0.00247	0.00149	0.06684	0.06015	0.8999
640	D	-0.00247	0.00149	0.06684	0.06010	0.8991
640	O	-0.00244	0.00150	0.06699	0.06006	0.8967
2560	F	-0.00083	0.00075	0.03338	0.03226	0.9664
2560	D	-0.00083	0.00075	0.03338	0.03227	0.9669
2560	O	-0.00088	0.00075	0.03355	0.03227	0.9620

Table S3.6. Coverage uncertainty and widths. Widths refer to the original Wald interval. These are coverage probabilities, not null rejection rates.

n	Method	Coverage	MCSE	95% Wilson	Mean width
160	F	0.8580	0.00780	[0.8420, 0.8726]	0.4090
160	D	0.8535	0.00791	[0.8373, 0.8683]	0.4054
160	O	0.8485	0.00802	[0.8321, 0.8635]	0.4052
640	F	0.9115	0.00635	[0.8983, 0.9232]	0.2358
640	D	0.9110	0.00637	[0.8977, 0.9227]	0.2356
640	O	0.9050	0.00656	[0.8914, 0.9171]	0.2354
2560	F	0.9270	0.00582	[0.9148, 0.9376]	0.1264
2560	D	0.9265	0.00584	[0.9142, 0.9371]	0.1265
2560	O	0.9290	0.00574	[0.9169, 0.9395]	0.1265

Table S3.7. Paired complete-minus-direct coverage.

n	Paired R	Coverage difference	Paired MCSE	Full only	Direct only
160	2000	0.0045	0.001800	11	2
640	2000	0.0005	0.000500	1	0
2560	2000	0.0005	0.000866	2	1

S3.5 Supporting C and marginal profile CVs

The saved pathway records include each fitted C and rho but not both marginal CVs. For the existing tau=.10,n=640 cell only, the integration script deterministically regenerates each of the 2,000 original arrays using its recorded seed mapping and evaluates distances at the *stored* references. It performs no reference refitting, simulation expansion or rejection recalibration. It checks all 8,000 stage records against saved C and rho and verifies $C=1+\rho CV_x CV_y$ per replication before summarizing.

Table S3.8. Secondary C and CV summaries. Each entry is mean (MCSE). The three means do not reconstruct mean C when multiplied by mean rho.

Stage	C (MCSE)	CV _x (MCSE)	CV _y (MCSE)
R	0.999994 (0.000009)	0.100108 (0.000064)	0.100172 (0.000062)
S	1.001573 (0.000051)	0.107351 (0.000223)	0.107407 (0.000223)
M	1.001573 (0.000051)	0.107351 (0.000223)	0.107407 (0.000223)
F	1.079550 (0.001963)	0.265546 (0.003200)	0.265044 (0.003204)

The CVs use divisor n, not n-1. C comes from the original stored `historical_C` field, which here denotes robust-reference C, not original all-to-all c-delta. The generated integration audit records maximum replay gaps, record counts and source hashes. This verifies the bounded summary; it is not a fresh Monte Carlo run or independent clean reproduction.

S3.6 Post hoc diagnostics and root flags

Table S3.9. Nonzero-skew tail and population-SE diagnostics. These use the same frozen replications. P=reported plug-in SE; V=infeasible population asymptotic SE. All rows use the complete estimator and 2,000 replications. Below means the interval is wholly below truth; above means it is wholly above truth. Wilson intervals are retained in the source file.

n	SE	Coverage	Below	Above	Coverage delta	Paired MCSE
160	P	0.8580	0.1115	0.0305	0.0000	0.00000
160	V	0.9675	0.0080	0.0245	0.1095	0.00750
640	P	0.9115	0.0695	0.0190	0.0000	0.00000
640	V	0.9590	0.0100	0.0310	0.0475	0.00618
2560	P	0.9270	0.0595	0.0135	0.0000	0.00000
2560	V	0.9580	0.0195	0.0225	0.0310	0.00525

The population-SE substitution changes random scale and its dependence on the estimate simultaneously. It is not a practical remedy or unique causal decomposition. Eighteen regular-study reference replays had maximum scale-normalized discrepancy about 2.35e-11; those fits do not suggest an obvious solver explanation for this undercoverage.

Table S3.10. The two flagged pathway full fits. Both are tau=.10,n=640. The flag compares stored and bracketed roots in fitted-scale units against 1e-6. Delta rho is bracketed minus original; relative delta SE is a fraction, not a percentage. Neither studentizer's rejection changed.

Rep	Root gap / scale	Delta rho	Relative delta SE	Decisions changed
70	1.410854e-05	-4.572481e-07	-3.02298e-05	0 / 0
1622	1.790506e-06	-4.681389e-07	-1.397699e-05	0 / 0

The recorded score residual at Stage 1 need not be zero: those references are fixed at population values and are not empirical roots. It must not be counted as a fitting failure. The public full-fit solver has not been silently replaced by the bracketed solver; no flagged record was removed.

S3.7 Permutation seed ledger and execution order

The full bridge DGM and methods are in S2. The four-family seed formula is 2026081750+n+round(10000*epsilon)+100000*f, with zero-based family order uniform, exponential, half-normal, scaled_beta12. Each replication consumes the draw, 99 permutation index vectors, then the historical diagnostic bootstrap calls (199 reference fits per margin) before the next draw. Pilot and larger confirmation use the same cell seed and call order, so the pilot is a prefix of the larger confirmation; they are not independent datasets to be pooled. For the prospective family the seed is 2026082581+100000*i+10000*j+n, with i=1,2 for n=80,320 and j=1,2,3 for epsilon=.05,.10,.20; the stream has no diagnostic bootstrap step.

Table S3.11. Exact permutation cell ledger. P=24-cell pilot, C=larger matched-index confirmation, H=prospective hyperexponential. All have 99 permutations per dataset. R is number of datasets.

Panel	Family	n	epsilon	R	Cell seed
P	Exp	80	0.05	150	2026182330
P	Half-normal	80	0.05	150	2026282330
P	Beta	80	0.05	150	2026382330
P	Uniform	80	0.05	150	2026082330
P	Exp	80	0.1	150	2026182830
P	Half-normal	80	0.1	150	2026282830
P	Beta	80	0.1	150	2026382830
P	Uniform	80	0.1	150	2026082830
P	Exp	80	0.2	150	2026183830
P	Half-normal	80	0.2	150	2026283830
P	Beta	80	0.2	150	2026383830
P	Uniform	80	0.2	150	2026083830
P	Exp	320	0.05	150	2026182570
P	Half-normal	320	0.05	150	2026282570
P	Beta	320	0.05	150	2026382570
P	Uniform	320	0.05	150	2026082570
P	Exp	320	0.1	150	2026183070
P	Half-normal	320	0.1	150	2026283070
P	Beta	320	0.1	150	2026383070
P	Uniform	320	0.1	150	2026083070
P	Exp	320	0.2	150	2026184070
P	Half-normal	320	0.2	150	2026284070
P	Beta	320	0.2	150	2026384070
P	Uniform	320	0.2	150	2026084070
C	Exp	320	0.05	500	2026182570
C	Half-normal	320	0.05	500	2026282570
C	Beta	320	0.05	500	2026382570
C	Uniform	320	0.05	500	2026082570
H	Hyperexp	80	0.05	200	2026192661
H	Hyperexp	80	0.1	200	2026202661
H	Hyperexp	80	0.2	200	2026212661
H	Hyperexp	320	0.05	200	2026292901
H	Hyperexp	320	0.1	200	2026302901
H	Hyperexp	320	0.2	200	2026312901

The code paths are `run_family_cell` in `scripts/run_profile_bridge_family_validation_20260817.py` and `claim3_prospective_family_validation` in `scripts/run_claim_external_validation_20260825.py`. The index calculations and original source links are preserved in the September 5 display tables.

S3.8 Source ledger and reproducibility status

All paths below are relative to the repository root. Raw and normalized SHA-256 hashes of these inputs are stored in the new integration audit, alongside the old manifest verification. The old result files and manifests are unchanged. An unexpected old hash failure must be investigated, not repaired by updating the expected hash.

ID	Used for
W	Main Table 1; S3.1
W0	Regular-null source/denominators
B	S2.1; legacy Figure 2
F	S2.2–3; legacy Figure 3
D	Legacy display source hashes
P	All 32,000 stage records
PS	Main Tables 3–4; S3.2–3
PP	S3.4 paired contrasts
PR	S3.10 retained root flags
PM	Pathway provenance
A	18,000 method records
AS	Main Table 2; S3.5–6
AP	S3.7 paired coverage
AT	Population truth and IF variance
AD	Contamination derivative check
AM	Skew provenance
Q	S3.9 post hoc tail diagnostics
QR	18 regular reference replays
QM	Post hoc provenance

- W: results/section6_display_table2_wald_20260905.tsv.
- W0: results/claim1_wald_validation_20260823.tsv.
- B: results/section6_display_figure2_bridge_20260905.tsv.
- F: results/section6_display_figure3_residual_20260905.tsv.
- D: results/section6_display_manifest_20260905.tsv.
- P: results/reference_pathway_replications_20260909.tsv.
- PS: results/reference_pathway_summary_20260909.tsv.
- PP: results/reference_pathway_paired_20260909.tsv.
- PR: results/reference_pathway_root_sensitivity_20260909.tsv.
- PM: results/reference_pathway_manifest_20260909.tsv.
- A: results/active_nuisance_replications_20260907.tsv.
- AS: results/active_nuisance_summary_20260907.tsv.
- AP: results/active_nuisance_paired_20260907.tsv.
- AT: results/active_nuisance_population_20260907.tsv.
- AD: results/active_nuisance_derivative_20260907.tsv.
- AM: results/active_nuisance_manifest_20260907.tsv.
- Q: results/referee_coverage_diagnostics_20260908.tsv.
- QR: results/referee_root_replay_20260908.tsv.
- QM: results/referee_diagnostics_manifest_20260908.tsv.

The deterministic table/replay check is `python scripts/integrate_manuscript_20260911.py` from the repository root. Its default mode is read-only and rejects changed displayed tables. Only `--write` refreshes the new manuscript table blocks and dated integration report. It never rewrites frozen scientific sources. The older completion audit remains a separate check of the historical chapter files.

For a future clean reproduction, use an isolated checkout, an explicit environment lock and a separate output directory. The September 7 script accepts `--out-dir`; older runners write to fixed result paths and therefore must only be run in that isolated checkout. Reproduction must compare declared source versions, numeric tolerances, row accounting and retained failures, not merely whether a script exits successfully. No clean-environment Monte Carlo reproduction is claimed by the present table assembly.

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